Le Xu

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I'm currently a researcher/software engineer at Computing Infrastructure Lab of Bytedance Inc. My current work focuses on building next-generation AI serving infrastructures. Previously I was a research fellow of Computer Science at UT Austin, working with Prof. Aditya Akella at UT Networked Systems group (UTNS). I was one of the recipients of CIFellow 2021, supported by both National Science Foundation (NSF) and Computing Research Association (CRA). During my PhD I worked on making real time data processing systems more elastic, under the supervision of Prof. Indranil Gupta at Distributed Protocols Research Group (DPRG). My current research mainly revolves around all aspects of resource-efficient cloud computing frameworks, including serverless computing and alternative system architectures. The list of my publication could be found here: https://scholar.google.com/citations?user=uFyE-AQAAAAJ&hl=en.

Education

08/2015–12/2021 University of Illinois at Urbana-Champaign
Ph.D. in Computer Science (advised by Indranil Gupta)
Thesis: "Elastic techniques to handle dynamism in real-time data processing systems."

08/2013–05/2015 University of Illinois at Urbana-Champaign
M.S in Computer Science (advised by Indranil Gupta)
Thesis: "Stela: on-demand elasticity in distributed data stream processing systems"

08/2009–05/2013 University of Illinois at Urbana-Champaign
B.S in Math and Computer Science

Selected Work Experience

04/2024–Present Computing Infrastructure Lab, Bytedance Inc. (Researcher/Software Engineer)
Took the leadership role in project AIBrix, a scalable, cloud-native infrastructure for deploying large language models (LLMs), which supports production services at Bytedance and has gained significant community adoption as an open-source project (3700+ GitHub stars). My work focuses on redesigning LLM serving infrastructure to support performance isolation, diverse SLO guarantees, and KV-cache–aware routing by co-optimizing system components across the stack. This involves tight integration between AIBrix and external services to enable adaptive, efficient inference. I also developed a benchmarking suite informed by production workloads to evaluate system reliability and provide standardized comparisons across LLM serving frameworks.

09/2021–04/2024 University of Texas at Austin (Research Fellow)
I architected and implemented dynamic, adaptive serving frameworks for modern ML workloads—including MOSEL for multimodal inference, Dirigo for serverless stream processing, and BlockLLM for block-based multi-tenant LLM serving. My contributions span intelligent input modality selection, message-level autoscaling, performance isolation under SLOs, reusable model component orchestration, KV-cache coordination, and speculative execution. These systems achieved significant improvements in throughput, latency, resource utilization, and scalability across diverse deployment contexts.

06/2019-08/2019 Alibaba Damo Academy, Data Analytics and Intelligence Lab (Research Intern)
Architected Banyan, a subquery-aware dataflow engine for graph queries that achieved up to 1,000× speedup and robust multi-tenant isolation through fine-grained scheduling and dynamic load balancing.

05/2017-08/2017 Microsoft, Cloud and Information Services Lab (Research Intern)
Developed Chi, a scalable control plane for stream processing that embeds reactive control logic in data channels to enable low-latency scheduling, checkpointing, and reconfiguration across distributed operators.

05/2016-08/2016 Hewlett-Parkard Labs, Software Analytics Group (Research Intern)
Conducted performance modeling and analysis Sparkle, a shared-memory optimization library for Apache Spark that replaced TCP/IP shuffles with off-heap memory stores, yielding up to 20× speedups and improved GC efficiency in graph analytics

Publication

- Daoxuan Xu, Le Xu, Jie Ren, Yifan Sun. "Exploring the Wafer-Scale GPU." Workshop on General Purpose Processing using GPU (GPGPU), 2025.
- Bodun Hu, Le Xu, Jeongyoon Moon, Neeraja J. Yadwadkar, Aditya Akella. "MOSEL: Inference Serving Using Dynamic Modality Selection." The 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2024.
- Peter Schafhalter, Sukrit Kalra, Le Xu, Joseph E. Gonzalez, and Ion Stoica. "Leveraging Cloud Computing to Make Autonomous Vehicles Safer." IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2023.
- Li Su, Xiaoming Qin, Zichao Zhang, Rui Yang, Le Xu, Indranil Gupta, Wenyuan Yu, Kai Zeng, Jingren Zhou "Banyan: A Scoped Dataflow Engine for Graph Query Service." Proceedings of the VLDB Endowment (VLDB), 2022.
- Le Xu, Shivaram Venkataraman, Indranil Gupta, Luo Mai, Rahul Potharaju. "Move Fast and Meet Deadlines: Fine-grained Real-time Stream Processing with Cameo" USENIX Conference on Networked Systems Design and Implementation (NSDI), 2021. *(acceptance rate 16.6%)*
- Yunhui Long, Le Xu, Carl Gunter "A Hypothesis Testing Approach to Sharing Logs with Confidence" ACM Conference on Data and Application Security and Privacy (CODASPY), 2020.
- Faria Kalim, Le Xu, Sharanya Bathey, Richa Meherwal, Indranil Gupta. "Henge: Intent-driven Multi-Tenant Stream Processing" Proceedings of the ACM Symposium on Cloud Computing (SoCC), 2018. *(acceptance rate 24.3%)*
- Luo Mai, Kai Zeng, Rahul Potharaju, Le Xu, Steve Suh, Shivaram Venkataraman, Paolo Costa, Terry Kim, Saravanan Muthukrishnan, Vamsi Kuppa, Sudheer Dhulipalla, Sriram Rao. "Chi: a scalable and programmable control plane for distributed stream processing systems." Proceedings of the VLDB Endowment (VLDB), 2018.
- Mainak Ghosh, Ashwini Raina, Le Xu, Xiaoyao Qian, Indranil Gupta, Himanshu Gupta. "Popular is Cheaper: Curtailing Memory Costs in Interactive Analytics Engines." Proceedings of the 2018 European Conference on Computer Systems, (EuroSys), 2018. *(acceptance rate 16.5%)*
- Mainak Ghosh, Le Xu, Indranil Gupta. "Resource Management: Performance Assuredness in Distributed Cloud Computing via Online Reconfigurations" Assured Cloud Computing. John Wiley & Sons, 2018.
- Mijung Kim, Jun Li, Haris Volos, Manish Marwah, Alexander Ulanov, Kimberly Keeton, Lucy Cherkasova, Le Xu, Pradeep Fernando. "Sparkle: Optimizing Spark for Large Memory Machines and Analytics" 2017 ACM Symposium on Cloud Computing (SOCC), poster track. 2017.
- Le Xu, Boyang Peng, and Indranil Gupta. "Stela: Enabling Stream Processing Systems to Scale-in and Scale-out On-demand." *IEEE International Conference on Cloud Engineering (IC2E)*. 2016. *(acceptance rate 21.9%)*
- Wenting Wang, Le Xu, and Indranil Gupta. "Scale Up vs. Scale Out in Cloud Storage and Graph Processing Systems." *2015 IEEE International Conference on Cloud Engineering (IC2E)*. IEEE, 2015.

Preprint

- Jiaxin Shan, Varun Gupta, Le Xu, Haiyang Shi, Jingyuan Zhang, Ning Wang, Linhui Xu, Rong Kang, Tongping Liu, Yifei Zhang, Yiqing Zhu, Shuowei Jin, Gangmuk Lim, Binbin Chen, Zuzhi Chen, Xiao Liu, Xin Chen, Kante Yin, Chak-Pong Chung, Chenyu Jiang, Yicheng Lu, Jianjun Chen, Caixue Lin, Wu Xiang, Rui Shi, Liguang Xie. "AIBrix: Towards Scalable, Cost-Effective Large Language Model Inference Infrastructure", 2025
- Jiamin Li, Le Xu, Hong Xu, Aditya Akella. "BlockLLM: Multi-tenant Finer-grained Serving for Large Language Models", 2024
- Le Xu, Divyanshu Saxena, Neeraja J. Yadwadkar, Aditya Akella and Indranil Gupta. "Dirigo: Self-scaling Stateful Actors For Serverless Real-time Data Processing.", 2023.

Book Chapter

- Ghosh, Mainak, Le Xu, and Indranil Gupta. "Resource Management: Performance Assuredness in Distributed Cloud Computing via Online Reconfigurations." Assured Cloud Computing (2018): 160-236.

Patent

- Potharaju, Rahul, Kai Zeng, Paolo Costa, Terry Yumin Kim, Sudheer Dhulipalla, Saravanan Muthukrishnan, Shivaram Venkataraman, Le Xu, M. A. I. Lao, and Sriram Rao. "Dataflow execution graph modification using intermediate graph." U.S. Patent Application 15/994,918, filed December 5, 2019.

Talk

- "Stela: Enabling Stream Processing Systems to Scale-in and Scale-out On-demand." IC2E'16.
- "Move Fast and Meet Deadlines: Fine-grained Real-time Stream Processing with Cameo" NSDI'21.
- "Dirigo: Self-scaling Stateful Actors For Serverless Real-time Data Processing." NITRD 30TH Anniversary Symposium, 2022.
- "Introducing AIbrix: A Testbed for Large-Scale LLM Inference System Research", ASPLOS'25 Tutorial.

Teaching Experience

- Cloud Computing Concepts (Coursera) - Teaching Assistant (01/2015, 01/2016)
- Advanced Distributed Systems - Teaching Assistant (01/2016)
- Distributed System - Teaching Assistant (01/2015, 09/2019)
- Cloud Computing Capstone (05/2018, 05/2020)

Awards and Honors

- 2021 CRA/CCC Computing Innovation Fellowship (CIFellows)
- 2020 EECS Rising Stars
- Tapia 2020 Conference Scholarship
- 2016 David J. Kuck Outstanding M.S. Thesis Award
- 2015 Outstanding Teaching Assistant
- Travel Grant: SOSP 2019, OSDI 2018, SoCC 2018, SOSP 2017, Grace Hopper Celebration 2015, IWCA 2015

Services

- Program Comittee: Middleware'23, NSDI'24, NSDI'26
- Reviewers: ACM/IFIP/USENIX International Middleware Conference, USENIX Symposium on Networked Systems Design and Implementation (NSDI), ACM Proceedings of Very Large Data Base Endowment, IEEE Transactions on Computers (TC), IEEE Transactions on Parallel and Distributed Systems (TPDS), IEEE International Symposium on Cluster, Cloud and Internet Computing (CCGrid), Transactions on Mobile Computing (TMC).

June 16, 2025